CSE-2 – Baseline RSL Prediction Model

ENG499 Cold Chain | Week 2/3 dataset task | Files: ml/rsl_training_table.csv, ml/lab_quality_checks.csv

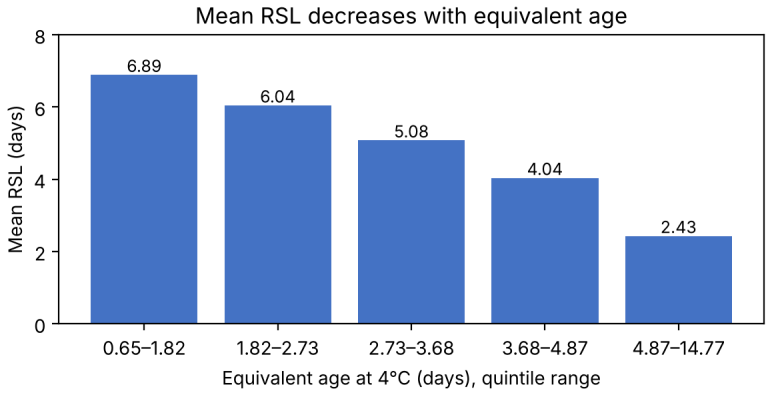
1. Objective

Build a simple, reproducible baseline that predicts target_rsl_days (remaining shelf life, RSL, in days) from a single feature, equiv_age_at_4c_days. This feature converts the observed temperature history into an equivalent number of days at the 4°C reference temperature, so a larger value should mean less RSL.

2. Data exploration

The training table has 12,053 rows from 217 lots. The target RSL has mean 4.897 days (std 1.890), median 5.039 days, 10th–90th percentile range 2.333–7.238 days, and a maximum of 10.463 days. Some rows reach 0, meaning the product is already at the end of its shelf life.

The correlation between equivalent age and RSL is −0.855, a strong negative relationship. Splitting rows into five equal-size groups by equivalent age shows RSL falling steadily from 6.89 to 2.43 days:



3. Model and validation design

Model: ordinary least-squares Linear Regression with one input feature:

predicted_RSL_days = 8.1542 − 0.9603 × equiv_age_at_4c_days

Each additional equivalent day at 4°C reduces predicted RSL by about 0.96 days; a product with zero equivalent age is predicted to have about 8.15 days.

To avoid leakage, all splits are grouped by lot_id, so rows from one production lot never appear in both training and test data. Two checks were run: (a) a single hold-out split with GroupShuffleSplit (80/20, random_state=42): 9,560 training rows from 173 lots and 2,493 test rows from 44 lots, with an explicit check that no lot appears in both; (b) 5-fold GroupKFold by lot_id to show how stable the result is.

4. Results on the test set

Validation	MAE (days)	RMSE (days)	R^2
Hold-out test (44 lots)	0.865	1.158	0.657
5-fold GroupKFold (mean ± std)	0.765 ± 0.039	0.979 ± 0.067	0.731 ± 0.030

MAE 0.865 means predictions are off by about 0.87 days on average; RMSE is larger because big errors are weighted more. R^2 = 0.657 means the model explains about 66% of RSL variation on the hold-out lots. The hold-out split is somewhat harder than the average fold (MAE 0.865 vs 0.765), so the cross-validated figures are the more representative estimate, and we report both.

5. Evaluation against laboratory checks

lab_quality_checks.csv has 154 plate-count samples (spc_log_cfu_ml, about ±0.25 log measurement error). It does not contain RSL in days, so we do not compute day-based MAE/RMSE by comparing RSL with bacterial counts. Of the 154 samples, 153 were matched to a training row by su_id and nearest time. We used the 20,000 cfu/mL limit (log10 = 4.30) for a classification check: does the model predict RSL ≤ 0 when the lab count is above the limit?

Check	Result	
Lab samples matched	153 of 154	
Samples above the limit (lab)	3	
Samples the model predicts as expired (RSL ≤ 0)	0	
Agreement (above/below limit)	98.0% (150/153)	

Interpretation: the 98% agreement is misleading. Only 3 samples exceed the limit and the model flagged none of them, so it agrees only because it predicted “not expired” for nearly everything. This is a real limitation: the baseline does not detect the rare spoiled samples. The lab check is therefore weak evidence for or against the model. A day-based lab error requires either a growth rate (log10 cfu/mL per day) from the project’s growth/lag model, which can be set in the script, or reference RSL labels at the sample times.

6. Limitations and next steps

- One feature and a linear relationship: temperature-history details (e.g. time above a threshold, lag phase) and lot or store effects are not captured.
- Inspect residuals by checkpoint, store and lot (cse2_residuals_by_group.csv) for systematic over- or under-prediction, especially near RSL = 0.
- Define the plate-count-to-RSL conversion with IE-1 / the growth model, then add a day-based lab metric.
- Compare with a non-linear model (e.g. gradient boosting) using the same lot-grouped validation.
- Interface with IE-2: output field rsl_predicted_days, unit days, at the relevant dispatch or delivery time.

Reproducibility

Code: cse2_rsl_baseline.py (fixed random seed 42). Run from the dataset root; it writes cse2_test_predictions.csv, cse2_baseline_results.json and cse2_residuals_by_group.csv. All numbers above come from this run on the project dataset.